Sugarcane Variety Classifier Technical Documentation

This document compares the ResNet18 and YOLOv8 pipelines using the currently configured artifact folders. It summarizes training outcomes, validation and test performance, deployment-ready outputs, and any generated visual evidence stored in the artifact tree.

Model Comparison Overview

Model	Status	Best Val Acc	Test Acc	Artifact Classes	Artifacts Directory
resnet18	loaded	97.69%	97.54%	9	/home/u0_a387/viscane_training/content/data/sugarcane_artifacts/resnet18
yolov8	loaded	98.18%	97.68%	9	/home/u0_a387/viscane_training/content/data/sugarcane_artifacts/yolov8

Dataset and Image Summary

This section describes the prepared image dataset used for the documented training run. Counts are loaded from /home/u0_a387/viscane_training/content/data/prepared/prepared_dataset_analysis.json.

Dataset Configuration

Raw Directory	content/data/raw/DATASETSFINAL
Prepared Directory	/home/u0_a387/viscane_training/content/data
Label Mode	variety_maturity
Total Images	10,229
Class Count	9
Class Balance Ratio	50.14%
Preprocess Resize	384
Model Image Size	320

Split Totals

Dataset Section	Number of Images
Training	3,416
Validation	3,404
Testing	3,409

Class Image Counts

Class	Variety	Maturity	Training	Validation	Testing	Total
524__MATURE	524	MATURE	329	327	328	984
524__NOT_MATURE	524	NOT_MATURE	357	356	357	1,070

Class	Variety	Maturity	Training	Validation	Testing	Total
524__OVER_MATURE	524	OVER_MATURE	232	231	231	694
847__MATURE	847	MATURE	436	434	433	1,303
847__NOT_MATURE	847	NOT_MATURE	421	420	421	1,262
847__OVER_MATURE	847	OVER_MATURE	462	461	461	1,384
Mauritio__MATURE	Mauritio	MATURE	442	440	441	1,323
Mauritio__NOT_MATURE	Mauritio	NOT_MATURE	381	380	381	1,142
Mauritio__OVER_MATURE	Mauritio	OVER_MATURE	356	355	356	1,067

Variety Rollup

Variety	Training	Validation	Testing	Total
524	918	914	916	2,748
847	1,319	1,315	1,315	3,949
Mauritio	1,179	1,175	1,178	3,532

Maturity Rollup

Maturity	Training	Validation	Testing	Total
MATURE	1,207	1,201	1,202	3,610
NOT_MATURE	1,159	1,156	1,159	3,474
OVER_MATURE	1,050	1,047	1,048	3,145

Low-Sample Warnings

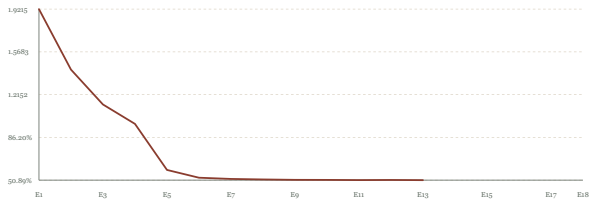
No low-sample warnings were recorded for the configured threshold.

Cross-Model Training and Validation Comparison

These shared charts normalize saved ResNet18 and YOLOv8 epoch history into comparable train/validation loss and accuracy views. YOLO falls back to results.csv when JSON epoch history is not available.

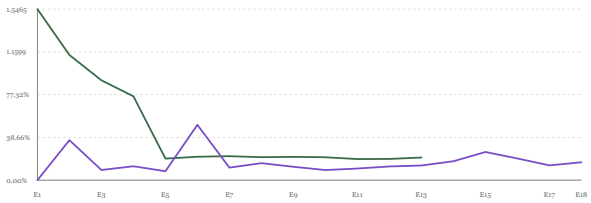
Combined Training Loss

ResNet18 Train Loss



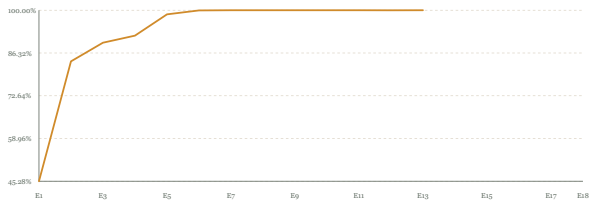
Combined Validation Loss

ResNet18 Validation Loss YOLOv8 Validation Loss



Combined Training Accuracy

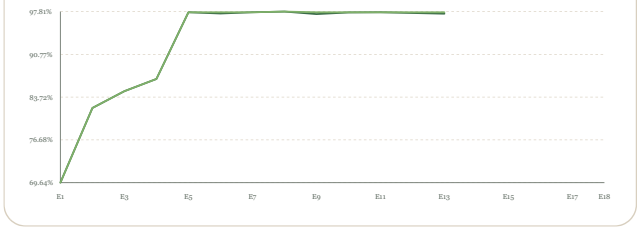
ResNet18 Train Accuracy



Combined Validation Accuracy

ResNet18 Validation Accuracy

ResNet18 Best Validation Accuracy



RESNET18

Loaded

Checkpoint: /home/u0_a387/viscane_training/content/data/sugarcane_artifacts/resnet18/best_model.pt

Artifacts Directory: /home/u0_a387/viscane_training/content/data/sugarcane_artifacts/resnet18

Load Error: none

Training Summary

Best Validation Accuracy	97.69%
Test Accuracy	97.55%
Epochs	45
Batch Size	32
Learning Rate	0.02%
Weight Decay	0.10%
Image Size	320

Evaluation Summary

Test Accuracy	97.54%
Variety Accuracy	99.88%
Maturity Accuracy	97.54%
Samples	3409
Friendly Outcome	Excellent result. The model is highly reliable on this test set and is a strong candidate for pilot deployment.

Android / Deployment Metadata

Android Artifact	/home/u0_a387/viscane_train
ONNX Artifact	/home/u0_a387/viscane_train
Image Size	320

Training Configuration

Device	cuda
Batch Size	32
Epochs	45
Learning Rate	0.02%
Weight Decay	0.10%
Seed	42

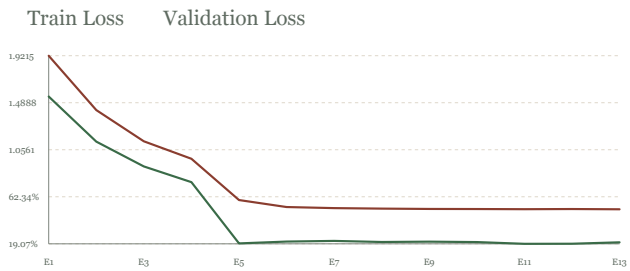
Augmentation and Robustness Settings

Validation Augmented	False
Train Resize Size	256
Train Crop Size	320
Validation Resize Size	320
Test Resize Size	320
Noise Std	0.00%

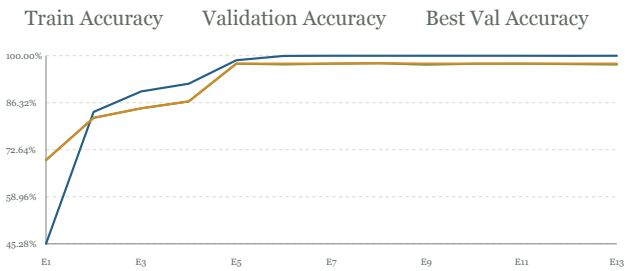
Blur Probability	0.00%
Erase Probability	0.00%
Rotation Degrees	3.0000
Crop Scale	2
Crop Ratio	2
Horizontal Flip Probability	30.00%
Brightness Range	2
Contrast Range	2
Saturation Range	2
Hue Range	2

Training Curves

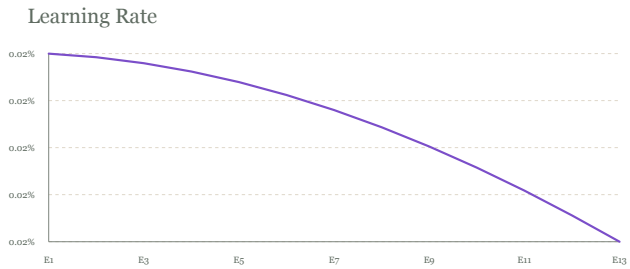
Loss Across Epochs



Accuracy Across Epochs



Learning Rate Schedule



Training and Validation History

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Best Val Acc	Learning Rate
1	1.9215	45.28%	1.5465	69.64%	69.64%	0.02%

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Best Val Acc	Learning Rate
2	1.4216	83.65%	1.1314	81.93%	81.93%	0.02%
3	1.1341	89.60%	90.30%	84.70%	84.70%	0.02%
4	97.35%	91.87%	75.81%	86.71%	86.71%	0.02%
5	59.35%	98.69%	19.49%	97.69%	97.69%	0.02%
6	52.91%	99.94%	21.14%	97.49%	97.69%	0.02%
7	51.92%	100.00%	21.71%	97.69%	97.69%	0.02%
8	51.48%	100.00%	20.75%	97.81%	97.81%	0.02%
9	51.15%	100.00%	21.07%	97.40%	97.69%	0.02%
10	51.11%	100.00%	20.71%	97.66%	97.69%	0.02%
11	50.92%	100.00%	19.07%	97.69%	97.69%	0.02%
12	51.05%	99.97%	19.17%	97.58%	97.69%	0.02%
13	50.89%	100.00%	20.47%	97.46%	97.69%	0.02%

Per-Class Performance

Class	Accuracy	Support
524__MATURE	98.78%	328
524__NOT_MATURE	99.44%	357
524__OVER_MATURE	100.00%	231
847__MATURE	100.00%	433
847__NOT_MATURE	100.00%	421
847__OVER_MATURE	100.00%	461
Mauritio__MATURE	82.31%	441
Mauritio__NOT_MATURE	100.00%	381
Mauritio__OVER_MATURE	100.00%	356

Top Confusions

True Class	Predicted Class	Count
Mauritio__MATURE	Mauritio__NOT_MATURE	58
Mauritio__MATURE	Mauritio__OVER_MATURE	18
524__MATURE	524__NOT_MATURE	2
524__MATURE	847__NOT_MATURE	2
524__NOT_MATURE	524__MATURE	2

True Class	Predicted Class	Count
Mauritio__MATURE	524__OVER_MATURE	1
Mauritio__MATURE	847__NOT_MATURE	1

Artifact Images

No image assets were found in this artifact folder.

YOLOV8

Loaded

Checkpoint: /home/u0_a387/viscane_training/content/data/sugarcane_artifacts/yolov8/yolov8/weights/best.pt

Artifacts Directory: /home/u0_a387/viscane_training/content/data/sugarcane_artifacts/yolov8

Load Error: none

Training Summary

Best Validation Accuracy	98.18%
Test Accuracy	97.68%
Epochs	45
Batch Size	32
Learning Rate	0.02%
Weight Decay	0.10%
Image Size	320
Checkpoint Path	/home/u0_a387/viscane_t ov

Evaluation Summary

Test Accuracy	97.68%
Variety Accuracy	97.68%
Maturity Accuracy	97.68%
Samples	3409
Friendly Outcome	Excellent result. The model is highly reliable on this test set and is a strong candidate for pilot deployment.

Android / Deployment Metadata

Android Artifact	/home/u0_a387/viscane_train
ONNX Artifact	/home/u0_a387/viscane_train
Image Size	320

Training Configuration

Batch Size	32
Epochs	45
Learning Rate	0.02%
Weight Decay	0.10%
Seed	42

Augmentation and Robustness Settings

Translate	2.00%
Scale	10.00%
Shear	0.00%
Perspective	0.00%
Flip Left/Right	30.00%
Flip Up/Down	0.00%

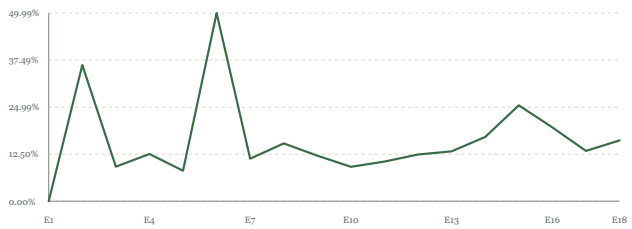
YOLO Base Weights	yolov8n-cls.pt

HSV Hue	0.10%
HSV Saturation	10.00%
HSV Value	10.00%
Mosaic	10.00%
MixUp	0.00%
Copy Paste	0.00%
Close Mosaic	10

Training Curves

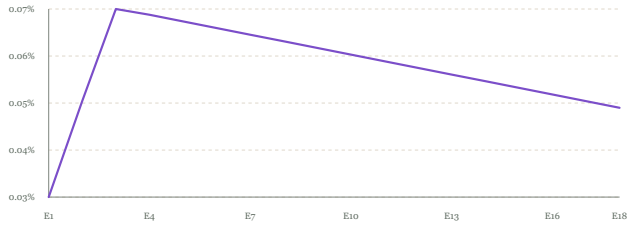
Loss Across Epochs

Validation Loss



Learning Rate Schedule

Learning Rate



No numeric chart series were found in the saved epoch history.

Training and Validation History

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Best Val Acc	Learning Rate
1	n/a	n/a	0.00%	n/a	n/a	0.03%
2	n/a	n/a	36.18%	n/a	n/a	0.05%
3	n/a	n/a	9.18%	n/a	n/a	0.07%
4	n/a	n/a	12.55%	n/a	n/a	0.07%
5	n/a	n/a	8.09%	n/a	n/a	0.07%
6	n/a	n/a	49.99%	n/a	n/a	0.07%
7	n/a	n/a	11.31%	n/a	n/a	0.07%

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Best Val Acc	Learning Rate
8	n/a	n/a	15.35%	n/a	n/a	0.07%
9	n/a	n/a	12.10%	n/a	n/a	0.06%
10	n/a	n/a	9.14%	n/a	n/a	0.06%
11	n/a	n/a	10.52%	n/a	n/a	0.06%
12	n/a	n/a	12.41%	n/a	n/a	0.06%
13	n/a	n/a	13.25%	n/a	n/a	0.06%
14	n/a	n/a	17.07%	n/a	n/a	0.05%
15	n/a	n/a	25.48%	n/a	n/a	0.05%
16	n/a	n/a	19.60%	n/a	n/a	0.05%
17	n/a	n/a	13.37%	n/a	n/a	0.05%
18	n/a	n/a	16.14%	n/a	n/a	0.05%

Per-Class Performance

This model did not save per-class metrics.

Top Confusions

No confusion pairs were recorded for this model.

Artifact Images

yolov8/confusion_matrix.png

yolov8/confusion_matrix_norma

yolov8/results.png

yolov8/train_batch0.jpg

yolov8/train_batch1.jpg

yolov8/train_batch2.jpg

yolov8/val_batch0_labels.jpg

yolov8/val_batch0_pred.jpg

yolov8/val_batch1_labels.jpg

yolov8/val_batch1_pred.jpg

yolov8/val_batch2_labels.jpg

yolov8/val_batch2_pred.jpg